

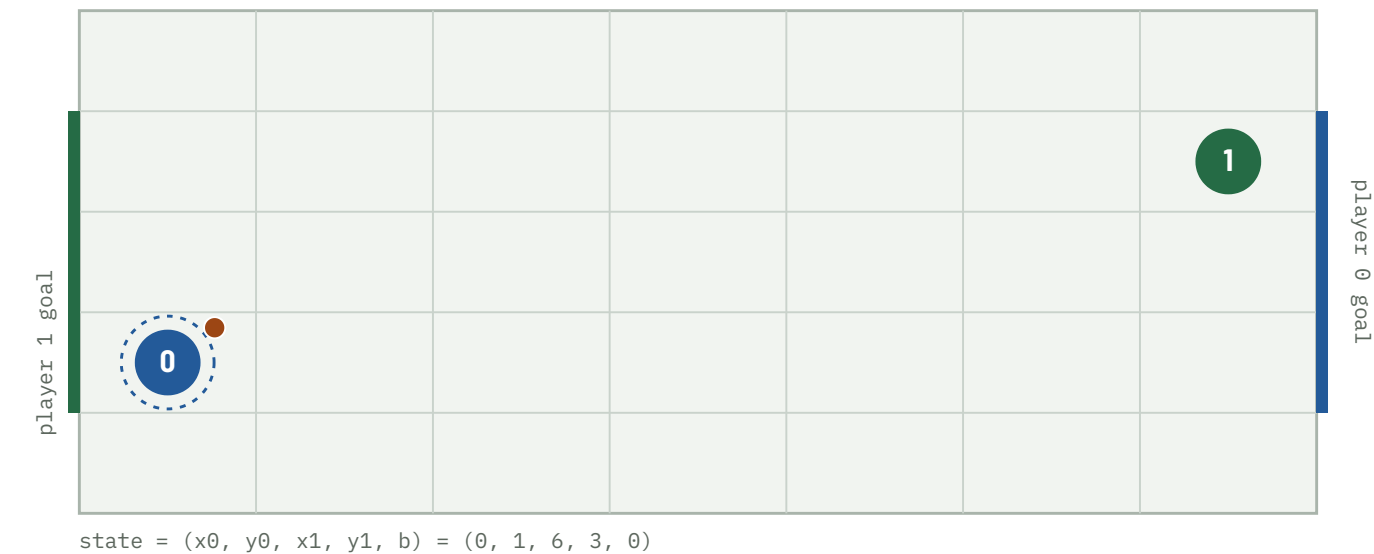
When Does Soccer Need Mixed Strategies?

A pure-first Nash-Q approach: where
a two-player soccer Markov game
does — and does not — need mixing,
and how much LP work that saves.

FOUR QUESTIONS

RQ1 Existence — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 210+ tests | game family in docs/design.md | methods in docs/methods.md | every policy drawn in docs/gallery.html



The kickoff. 7×5 grid, 2380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Start positions are the ones the A10 page assigns this netID. Actions U D L R are chosen simultaneously; reward is +1 / -1 / 0, zero-sum.

ABSTRACT

- 1** The deterministic A10 game — solved exactly, undiscounted, by 100-step backward induction — has a pure saddle at every one of its 238 000 (state $\times$ step) stage games. A pure equilibrium exists; the LP is never called; discounting was never load-bearing.

- 2** A coin-flip tie-break does not change this; Littman's random *move order* does — but only when the goal mouth is more than one cell wide, and even then on 3.95% of states (this configuration).

- 3** Those mixed states are geometrically localised: a decision tree predicts them from the state with precision 0.96, and 68 of 94 have a 2 $\times$ 2 matching-pennies support.

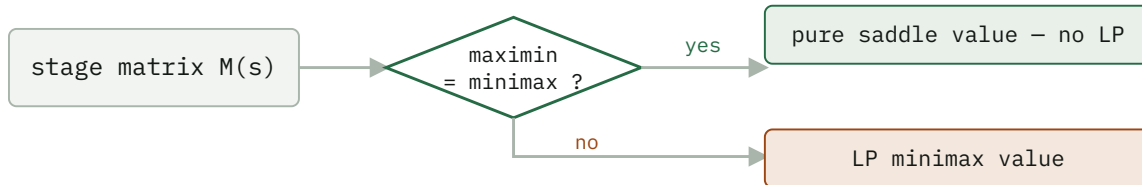
- 4** The pure-first hybrid solver matches the all-LP value function to 4e-16 while calling the LP on only 3.95% of states — $\sim 25\times$ fewer per sweep; mirror symmetry halves the work again.

Method

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s)[a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$\begin{aligned} V(s) = \text{val}(M(s)) &= \max_n \min_{k \in \mathcal{K}} (p^T M)[a_1] \\ &= \min_p \max_{a_0} (Mq)[a_0] \end{aligned}$$

The $Q = R + \beta \text{Nash}(Q')$ update is this operator. The A10 game is *undiscounted* (± 1 at a goal, tie after 100 steps); $\gamma < 1$ is a contraction device for value iteration and RQ4 shows the split holds across γ . Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via scipy HiGHS), and **hybrid** (the pure saddle value where maximin = minimax, the LP elsewhere — exact everywhere). Support enumeration (support_enum.py) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2380 states, so the LP is never called.

RQ1 – Existence

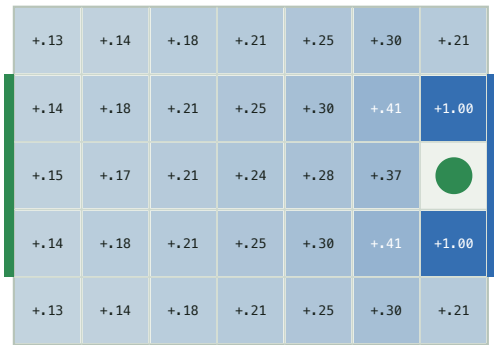
The A10 game is undiscounted, so it is solved exactly by 100-step backward induction (run_finite_horizon, experiments/undiscounted.csv):

EXACT UNDISCOUNTED GAME – 238 000 (STATE × STEP) STAGE GAMES

MOVE ORDER	MIXED STAGE GAMES	V(KICKOFF)	PURE EQUILIBRIUM?
deterministic (exact A10)	0	0.000	yes – pure (non-stationary)
coinflip tie-break	0	0.000	yes
random move order	3 148 (404 states)	+0.459	no

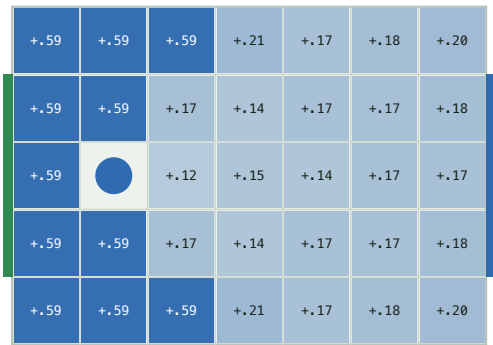
The deterministic equilibrium is **constructive and memoryless**: each player's win attractor (states from which it forces a goal, soccer_nash/attractor.py) equals its $V^* = \pm 1$ set, and the pure profile "attractor move on your winning set, safety move elsewhere" realizes V^* at every state (scripts/positional.py) — the game is positionally determined.

carrier position (player 0)



V^* by player 0 position (blue = player 0 ahead)

defender position (player 1)



V^* by player 1 position (blue = player 0 ahead)

The **value surface** of the random-order game ($\gamma = 0.9$). Left: V^* by carrier position – a smooth gradient toward the attacking goal, saturating to +1 on the two cells that score in one move. Right: V^* by defender position with the carrier held near midfield – nearly flat, the defender's location barely moves the value.

The stationary $\gamma < 1$ solve agrees and is faster — 0 no-pure-saddle stage games on the deterministic game at every γ from 0.5 to 0.995, and 94 / 2380 on the random game at $\gamma = 0.9$ ($V(\text{kickoff}) = +0.164$ at the netID start). The no-pure-saddle count never depends on the kickoff.

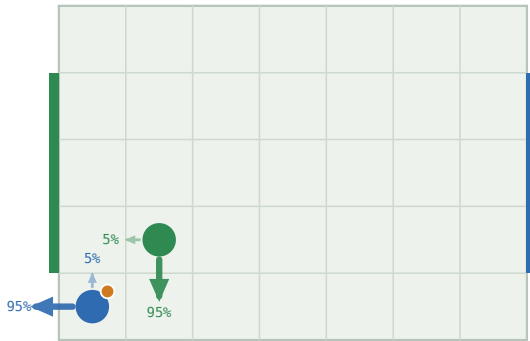
IT IS THE MOVE ORDER, NOT THE RANDOMNESS – AND IT HAS TO BE THE MAJORITY

The deterministic and coin-flip value functions are *identical* ($\max |V_{\text{det}} - V_{\text{coin}}| = 0$ at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames. Interpolating the two (move_order="blend", resolve by random order with probability blend) shows the onset is **sharp: 0** mixed stage games for $\text{blend} \leq 0.5$, dozens at $\text{blend} = 0.52$, and an *overshoot* to $\sim 1.5\times$ the fully-random count near $\text{blend} \approx 0.7$ before it relaxes (scripts/blend.py, docs/blend.md). A minority of random-order resolution is washed out by the deterministic tie-break.



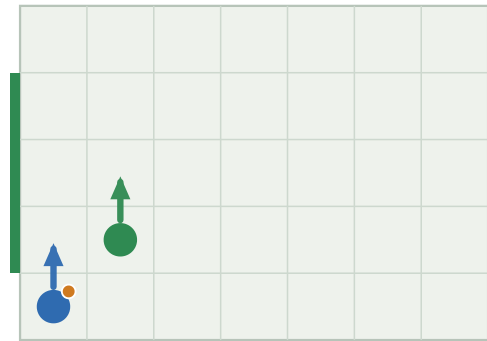
A threshold, not a ramp. No-pure-saddle stage games vs. the fraction of moves resolved by random order (5x4 board). Nothing until the random rule is the majority; then a jump and an overshoot.

random order -> mixed



mixed equilibrium · $V = +0.078$

deterministic -> pure



pure equilibrium · $V = +0.000$

One state, two resolution rules. Arrows are the equilibrium action distribution, sized by probability. Random order (Littman) makes the stage game matching pennies; deterministic carrier-wins collapses it to a pure best reply. Full set: [docs/gallery.html](https://docs.google.com/gallery).

RQ2 – Efficiency

Two numbers, kept apart: the **LP-call rate** is a property of the game and reproduces exactly; **wall clock** is a median of 5 repeats and depends on the machine and LP backend (`scripts/benchmark.py`).

RANDOM MOVE ORDER, $\Gamma = 0.9$ – THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	LP CALLS	STATES NEEDING LP	MEDIAN WALL CLOCK
pure (lower bound)	18	0	–	0.3 s
hybrid exact	108	9 672	3.95%	13.2 s
all-LP exact	108	~3.6 M	100%	~8 min

The hybrid solves an LP on only the 3.95% of states lacking a pure saddle — a $\sim 25\times$ per-sweep reduction — and matches the all-LP value to $4e-16$. The wall-clock ratio (~ 13 s vs ~ 8 min) is larger but less portable: it also reflects matrix construction and the LP backend. **Mirror symmetry** (`run_symmetric`, deterministic dynamics only): every state has a distinct board-flip-plus-player-swap mirror, so the 2380 states are 1190 pairs — reconstruct one from the other by $V(\text{mirror}(s)) = -V(s)$, cutting the deterministic solve 0.33 s $\rightarrow$ 0.23 s.

3.95% OF STATES, BUT 41% OF THE EQUILIBRIUM PATH

The LP-call rate counts the whole state space. Under the Nash policy from the kickoff only **456 of 2380** states are ever reached, and the 94 no-pure-saddle states carry **41%** of the discounted occupancy — a $2\times$ over-representation on the path, $10\times$ vs. the raw count (`soccer_nash/occupancy.py`, `docs/occupancy.md`). The carrier's optimal march to goal runs straight through the contested band; four of the eight most-visited states are mixed. The mixed structure is rare to *store* and common to *use*.

Freeze-then-iterate, and its staleness

RANDOM MOVE ORDER, HYBRID SOLVER – SAME FIXED POINT, $|\text{DIFF}| < 3E-9$

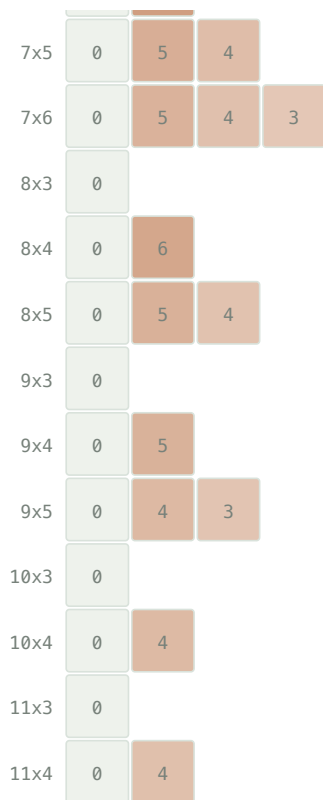
	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to $< 1e-8$. The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

RQ3 – Mechanism

Goal-mouth width, not board size, is the gate. The phase diagram sweeps every board with width ≤ 11 , height ≤ 9 , width $\cdot$ height ≤ 45 against goal widths 1 to height-2 (scripts/phase_diagram.py --analyze, experiments/phase_diagram.csv).

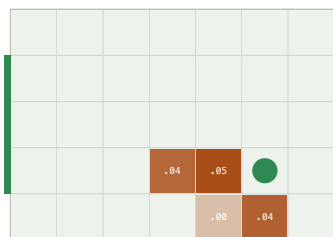
	goal-mouth width						
	1	2	3	4	5	6	7
3x3	0						
3x4	0	12					
3x5	0	10	9				
3x6	0	7	7	7			
3x7	0	7	6	6	6		
3x8	0	6	6	5	5	5	
3x9	0	6	5	5	4	4	4
4x3	0						
4x4	0	8					
4x5	0	8	6				
4x6	0	7	6	5			
4x7	0	7	5	5	4		
4x8	0	6	5	5	4	4	
4x9	0	6	5	4	4	4	3
5x3	0						
5x4	0	7					
5x5	0	6	4				
5x6	0	6	4	4			
5x7	0	5	4	4	3		
5x8	0	5	4	3	3	3	
5x9	0	5	4	3	3	3	3
6x3	0						
6x4	0	8					
6x5	0	6	4				
6x6	0	5	4	3			
6x7	0	4	4	4	3		
7x3	0						
7x4	0	7					



cells = mixed-state %

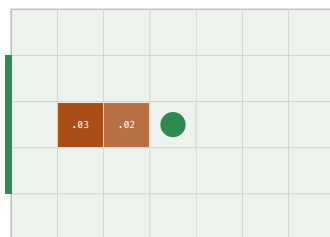
The gate. Goal width 1 → 0 mixed states, on all 40 one-cell configs. Goal width ≥ 2 → mixing on all 87: whether a board mixes is a *perfect* function of goal width. The *fraction* (2.7–12%) is an OLS R^2 0.64 in board area – a dilution effect – with goal width adding little and a negative coefficient. Every state is reachable from the kickoff.

defender at (5, 1)



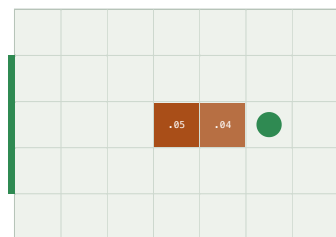
cells: minimax minus maximin of the carrier's stage game

defender at (3, 2)



cells: minimax minus maximin of the carrier's stage game

defender at (5, 2)



cells: minimax minus maximin of the carrier's stage game

Where the carrier has to guess. Defender pinned at the green disc; each cell shaded by minimax – maximin of the carrier's stage game (0 = pure saddle). Mixing is a local phenomenon – a handful of cells where the defender contests the forward lane.

Geometry predicts the label (scripts/geometry_model.py): a depth-4 decision tree on carrier-frame features separates mixed from the rest with **precision 0.96, recall 0.91**. The dominant feature is defender_can_intercept — the defender within one move of the carrier's forward cell ($P(\text{mixed}) = 0.44$ vs 0.002).

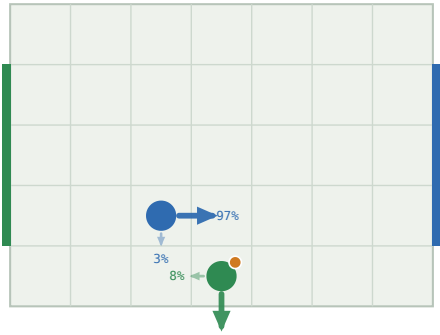
Beyond prediction, the 94 mixed states canonicalize under the board mirror to 47 pairs and cluster into 8 **geometric templates** (scripts/templates.py, docs/templates.md).

THE MIXED-STATE TEMPLATES – CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

STATES	GEOMETRY	SUPPORT	STRUCTURE
24	defender 1 ahead, carrier one row off the goal row	2×2	matching pennies
24	defender 1 ahead, same row, carrier on a goal row	2×2	matching pennies
16	defender 2 ahead, same row, carrier on a goal row	2×2	matching pennies
4	defender diagonally adjacent, carrier off the goal row	2×2	matching pennies
14	defender 1–2 ahead, same row, on a goal row	2×1	near-pure saddle
8	defender 2 ahead, same row, on a goal row	3×2	degenerate (row ties)
4	defender 2 ahead, carrier off the goal row	3×3	wider mix

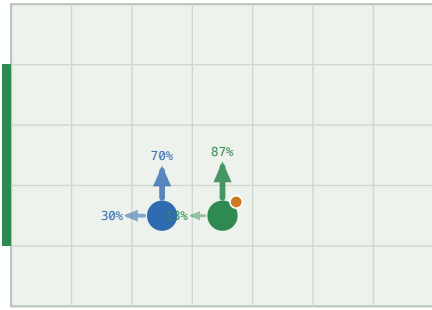
The four 2×2-support templates are all verified matching pennies — the carrier's best reply flips between the two defender columns and vice versa — covering 68 of 94 states: carrier {climb, advance} against defender {cover a lane, hold the forward cell}. All 94 share one value across every equilibrium; 64 have a unique one.

24 states $p^*=0.92$ $q^*=0.03$



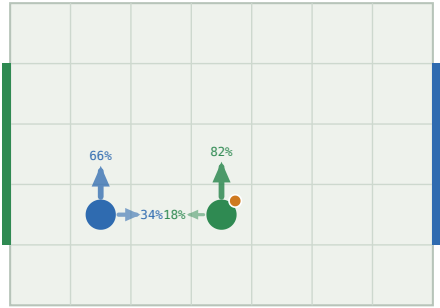
mixed equilibrium · $V = -0.150$

24 states $p^*=0.87$ $q^*=0.70$



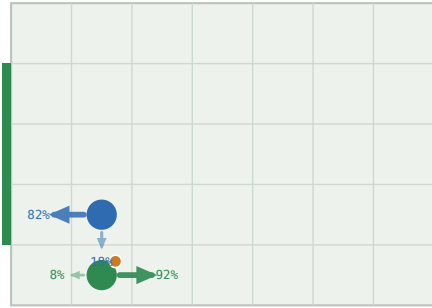
mixed equilibrium · $V = -0.199$

16 states $p^*=0.82$ $q^*=0.66$



mixed equilibrium · $V = -0.225$

4 states $p^*=0.08$ $q^*=0.18$



mixed equilibrium · $V = -0.206$

One policy fan per matching-pennies template. Each covers a class of mixed states related by carrier-frame geometry; p^* , q^* are the equilibrium weights on the first support action. The carrier randomizes between two scoring lanes, the defender between covering them.

Why the mix is unavoidable. A mixed stage game needs all three of: a stochastic ($\frac{1}{2}$ - $\frac{1}{2}$) resolution order; a goal mouth wider than one cell, so the carrier has two scoring lanes; and a defender close enough to contest the forward cell but unable to cover both lanes in one move. The carrier climbs or drops, the defender guesses which — matching pennies. Drop any one clause — a single goal cell, deterministic resolution, or the coin-flip tie-break — and every stage game has a pure saddle. The full argument and one stage matrix per template are in docs/templates.md. For the single goal cell the theorem is partly proved (docs/proof.md): the defender has a closed-form optimal strategy — guard the one cell — and every stage game is dominance-solvable, machine-checked for all boards up to 11×5 .

SHALLOW MIXES, BUT THEY COMPOUND

Every one of the 94 no-pure-saddle stage games sits within 0.07 of a pure saddle (minimax – maximin, median 0.029) — the equilibria are shallow, and the exact mixing weights are numerically delicate. What is robust: 68 have a 2×2 support and all 68 are structurally matching pennies, and the *value of mixing* — $V(\text{hybrid}) - V(\text{pure maximin})$ — is +0.13 on average at those states. It compounds along a path: at the centred kickoff a pure-strategy player secures only a draw (0.000) where mixing is worth +0.15 (scripts/mixing.py, docs/mixing.md).

The reward objective does not move the mixed region

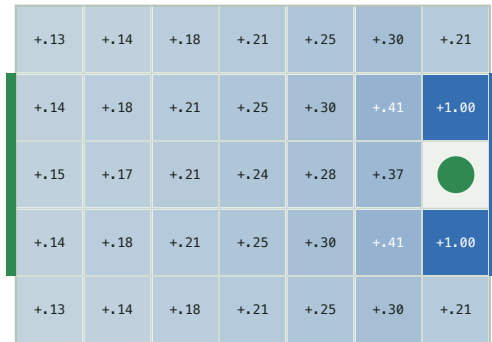
The default game rewards a *win* — first goal ends it, value $P(\text{win}) - P(\text{loss})$. A second objective, `scoring="rate"` (scripts/reward.py), keeps playing: a goal scores ± 1 and restarts with the ball to the conceding team, so the value is the expected discounted *goal difference* and $\gamma < 1$ is load-bearing. On the 7×5 random-order board:

SAME DYNAMICS, TWO REWARD OBJECTIVES – RANDOM MOVE ORDER, DISCOUNT 0.9

OBJECTIVE	NO-SADDLE STATES	VALUE RANGE	V(KICKOFF)
win – $P(\text{win}) - P(\text{loss})$	94	[−1.00, +1.00]	+0.150
rate – expected goal difference	94 – the same 94	[−0.88, +0.88]	+0.132

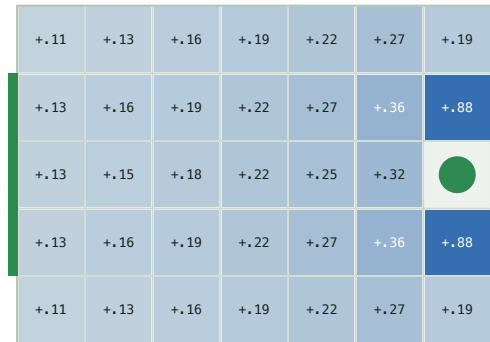
The no-pure-saddle set is *identical* — 0 states enter or leave. A mixed Nash equilibrium is an equilibrium of a single stage game $M(s)$; changing how the horizon is scored rescales $M(s)$ through $V(s')$ but does not cross the maximin = minimax boundary. The value *range* compresses ($\pm 1 \rightarrow \pm 0.88$): under rate a lost position is not a cliff — conceding restarts play with possession, a floor under every state. Details in docs/reward.md.

scoring = win ($P(\text{win}) - P(\text{loss})$)



V* by player 0 position (blue = player 0 ahead)

scoring = rate (expected goal difference)



V* by player 0 position (blue = player 0 ahead)

Same shape, compressed range. V* by carrier position under the two objectives. rate pulls the goal-adjacent +1 cells down to +0.88 – the concede-and-restart floor – without changing where the value gradient runs.

Littman's fifth action

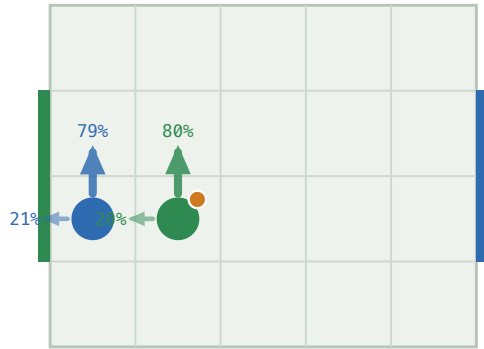
Littman's 1994 soccer game has a fifth action, stand, and his Figure 2 — a carrier pinned against its own goal by an adjacent defender — is the textbook "soccer needs mixing" example: the carrier must randomize between standing and moving. Adding stand to the random-order game (`n_actions=5`, `scripts/littman.py`) on Littman's 5×4 board:

LITTMAN'S BOARD (5×4, 2-CELL GOALS), RANDOM MOVE ORDER, DISCOUNT 0.9

ACTION SET	NO-SADDLE STATES	STAND IN THE MIXED SUPPORT	V(KICKOFF)
N, S, E, W	56	0	+0.147
N, S, E, W, stand	56	48	+0.172

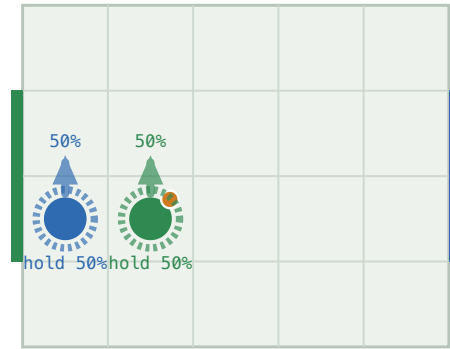
The fifth action does *not* move where mixing happens — 48 of the 56 no-pure-saddle states are shared, 8 swap in and 8 out at the margin — but it changes *what* the mix is: the carrier now hedges with stand in 48 of 56 states rather than with retreat, and every player is slightly better for the option ($V(\text{kickoff}) +0.147 \rightarrow +0.172$). The Figure 2 state is a clean matching-pennies matrix with equilibrium ($\frac{1}{2}$ climb, $\frac{1}{2}$ hold). The deterministic game keeps a pure saddle at every state with five actions too.

four moves



mixed equilibrium · $V = -0.255$

+ stand (Littman Fig 2)



mixed equilibrium · $V = -0.315$

Littman's Figure 2. Same state, four moves (left) vs. five (right). The dashed ring is stand. Adding it turns the carrier's climb/retreat hedge into climb/hold – Littman's mixed policy.

RQ4 – Numerical robustness

The value is three numbers

`value_bracket` returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^T Mq$), and can reach ($\max_i (Mq)_i$). The true value is bracketed, so the width certifies the LP error.

FINDING

With `scipy`'s HiGHS the three agree to $1.1e-16$ per stage game, $8.7e-10$ accumulated. The concern is real for older vertex/simplex solvers — a non-issue here.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
mixed states, $\gamma = 0.5 \rightarrow 0.9 \rightarrow 0.995$	0 → 0 → 0	122 → 94 → 102
classification split, <code>rel_tol</code> $1e-12 \rightarrow 1e-3$	stable	604 / 1682 / 94 (stable)
classification, <code>rel_tol</code> $1e-2 / 1e-1$	—	80 / 0 mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real minimax – maximin gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.

Secondary validation – the policy plays correctly

- Duality gap $V0_{br}(s_0) + V1_{br}(s_0) < 1e-9$ for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; $+0.162 \pm 0.002$ empirical discounted return over 5 seeds of 3000 games (vs. $+0.164$ computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability 0.43 — worse than random — which is the A10 competition's warning about non-equilibrium submissions.
- A10 networks over 5 seeds (experiments/a10_competition_seeds.csv): the Part 2 imitation net plays optimally at 0.990 ± 0.000 of states and wins Q8 every time; competition Network Second is exact, but Network First sits at a 0.19 ± 0.02 exploitability from this kickoff that does not wash out with re-seeding.

Beyond zero-sum, and open questions

- **General-sum — extension point.** `markov_game.py` solves 2-player general-sum Markov games by enumerating *all* stage equilibria (`support_enum.py`) and selecting one — the notes' "largest sum of values", a no-op for zero-sum but decisive for Battle of the Sexes, where it avoids the mixed equilibrium whose value is below either pure one. Pure-first throughout. The soccer game is zero-sum, so this does not touch the main result.
- **Function approximation — partial.** A network fit to the stage matrices (`nash_dqn.py`, $5 \rightarrow 96 \rightarrow 96 \rightarrow 16$ with biases, 600 epochs) trails the exact solver, over 5 seeds (`experiments/nash_dqn_seeds.csv`):

	EXACT HYBRID NASH-Q	NEURAL NASH-Q (MEAN $\pm$ SD)
max $ V - V_{\text{exact}} $	0	0.55 $\pm$ 0.02
mean $ V - V_{\text{exact}} $	0	0.13 $\pm$ 0.00
action agreement	100%	43% $\pm$ 2%
pure/mixed classification	100%	67% $\pm$ 2%
exploitability	$<1e-9$	0.44 $\pm$ 0.05
convergence (epochs to plateau, of 600)	exact	469 $\pm$ 31; 3/5 still creeping
runtime	9 sweeps, 0.3 s	600 epochs, ~35 s

Deterministic game only — `train_nash_dqn` rejects the stochastic variants, so the network never has to represent a genuinely mixed stage game. Even so it trails: it fits the ~1000 zero-value states easily (small *mean* error) but misses the γ^k bands and contested regions — worst-state error 0.55, policy about as exploitable as random play, mis-calls *whether* a state needs mixing a third of the time. Keep the exact solver as ground truth.

- **Four research questions** — the single-cell theorem's last step, whether the goal-width dichotomy is a known result, the move-order-vs-randomness distinction, and why freeze-then-iterate thrashes — are stated and answered as far as the evidence allows in `docs/discussion.md`.

Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Across the phase-diagram sweep every one-cell-goal board has 0 mixed states and every wider-goal board has some; the exact 3.95% is the 7×5 / 3-cell-goal case.
 - Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See docs/assumptions.md.
 - **Claim A/B** (a pure equilibrium of the deterministic game) is established constructively — the explicit pure memoryless attractor/safety profile. **Claim C** (the theoretical game necessarily has one, over all rules and settings) is *not*, for the general goal mouth; for the single goal cell it is partly proved (docs/proof.md).
 - random and coinflip are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.
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soccer_nash — game.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py ·
numerics.py · dominance.py · attractor.py · geometry.py · symmetry.py · tree.py ·
templates.py · onecell.py · reachability.py · occupancy.py · best_response.py · exploit.py
· mlp.py · nash_dqn.py · opponents.py · shaping.py · simulate.py · experiment.py ·
render.py · viz.py
scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py ·
templates.py · phase_diagram.py · onecell_proof.py · positional.py · benchmark.py ·
policy_iteration.py · selfplay.py · nash_dqn.py · render.py · gallery.py · littman.py ·
mixing.py · reward.py · blend.py · occupancy.py · a10_part1.py · a10_part2.py ·
a10_competition.py
docs — design.md · methods.md · discussion.md · assumptions.md · templates.md · proof.md ·
geometry.md · mechanism.md · littman.md · mixing.md · reward.md · blend.md · occupancy.md
· gallery.html
experiments — baseline.csv · undiscounted.csv · gamma_sweep.csv · tolerance_sweep.csv ·
board_sweep.csv · goal_mouth_sweep.csv · one_cell_goal.csv · phase_diagram.csv ·
nash_dqn_seeds.csv
env: 7×5 grid, 2380 states, zero-sum ± 1 , $\gamma = 0.9$